

CSX3007(541) Computer Architecture

Quiz II (2/2025)

This is a 3-hour online quiz (**10:00-13:00**), and students are not allowed to use any AI platforms, such as ChatGPT, Gemini, etc. to find answers (answers/formulas/equations obtained from such platforms are not valid for this exam). Use a scientific calculator instead of a calculator app. This is a **closed-book** examination (it carries a total of **70 points for 20 questions**)

* Indicates required question

1. Write your **Name, ID** and **Section** in the below space: *

1. (6 points) Assume that a **0.5 ns clock cycled** unpipelined processor executes a * benchmark program with the following instruction mix and instruction clock cycle count shown in the following Table. Calculate **effective CPI, MIPS rate, and Execution time (CPU time)**.

$$\text{CPI} = (4500 \times 3) + (3500 \times 2) + (2400 \times 4) + (1500 \times 5) / (4500 + 3500 + 2400 + 1500) = \mathbf{3.16}$$

$$\text{CPU time} = 3.16 \times 11900 \times 0.5 \times 10^{-9} = \mathbf{188800 \times 10^{-9} \text{ seconds}}$$

$$\text{MIPS rate} = \text{IC} / (\text{CPU time} \times 10^9) = 1 / (3.16 \times 0.5 \times 10^{-9} \times 10^6) = \mathbf{633}$$

Pic1

Instruction	Instruction Count	Clock cycle count
Integer arithmetic	4500	3
Data transfer	3500	2
Floating point	2400	4
Control transfer	1500	5

2. (1 point) Find the answer of the question that is shown in the Figure below: *

$$100 \times ((5-3)/3) = 66.7\%$$

Pic2

System **A** with 4.5 GHz speed performs **5 tasks per hour** while system **B** with a 4.5 GHz could complete only **3** tasks per hour. By what percentage is **A** faster than **B**?

3. (3 points) Find the answer of the question that is shown in the Figure below: *

$$w1(J1) = 0.167, w2(J2) = 0.222, w3(J3) = 0.278, w4(J4) = 0.333$$

$$\text{Weighted Harmonic mean} = 1/(0.167/0.5 + 0.222/0.45 + 0.278/0.53 + 0.333/0.43) \\ = 1/2.126 = 0.4704$$

Pic3

Consider **four jobs** being run on a corporate computer. The observed job execution rates per hour were **0.5, 0.45, 0.53**, and **0.43**. For a particular benchmark, **job1** is executed **30** times, **job2** is executed **40** times, **job3** is executed **50** times, and **job4** is executed **60** times. What is the *central tendency* of these measurements in jobs per hour?

4. (3 points) Assume that two computers (**A** and **B**) execute **four loops** of a scientific program. The details of their clock cycles for each loop are shown in the **Table** below. What is the central tendency in the speedup of the computers (from A to B)? *

$$\text{Ratio1} = 1.33, \text{Ratio2} = 2.5, \text{Ratio3} = 1.2, \text{Ratio4} = 1.6$$

$$\text{Geometric Mean} = \text{4th Square_root_of}(1.33 \times 2.5 \times 1.2 \times 1.6) = (6.384)^{0.25} = 1.59$$

Pic4

Loop	Comp A	Comp B
1	40	30
2	50	20
3	30	25
4	24	15

5. (4 points) Find the answer of the question that is shown in the Figure below: *

$$\text{Speed up} = 1/((1-a)+a/N)$$

P4600: $a = 0.6$, $N = 8$, Speedup = **2.11 times**

H6600: $a = 0.7$, $N = 8$, Speedup = **2.58 times**

H6600 is better in terms of efficiency.

Pic5

A benchmark test on two server systems, P4600 (an 8-core system) and H6600 (an 8-core system), has produced the following results: the P4600 shows 40% overhead during the test, whereas the H6600 has demonstrated the best performance during 70% of its test time. So, which system is more efficient for running your task (show your complete calculation steps)?

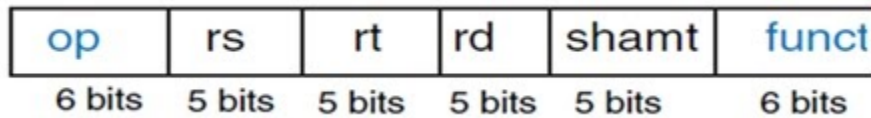
6. (1 point) Which one of the **four state elements** of the MIPS system support ID and WB stages? *

Register file (RF)

7. (6 points) Answer the following based on the instruction format given in the **Figure** below: (a). (1 point) What is the instruction type (in terms of the operands)? (b). (1 point) Show an example instruction (in assembly code). (c). (1 point) How many opcodes are possible? (d). (1 point) How many registers are in the system? (e). (1 point) How many functions are in the system? (f). (1 point) What is the size of the CPU? *

(a). **R-type** (b). **ADD R1, R2, R3** (c). **64** (d). **32** (e). **64** (f). **32-bit**

Pic6



8. (7 points) Answer the following questions based on the single-cycle data path of the MIPS system shown in the **Figure** below. Assume that the registers **R0** and **R1** have initialized with **0x00000007** and **0xC123FAE7**: (a). (1 point) Show the **I/P of the Sign Extend unit** (in hexadecimal). (b). 8.1.(1 point) Show the **O/P of the Sign Extend unit** (in hexadecimal). (c). (1 point) Show the **O/P of the ALU** (in hexadecimal). (d). (4 points) Show the result of the SW operation in Data Memory (each byte with its complete byte address).

(a). (1 point) **0x0008**

(b). (1 point) **0x00000008**

(c). (1 point) **0x0000000F**

(d). (4 points)

memory address: byte

0x0000000F : 0xC1

0x00000010 : 0x23

0x00000011 : 0xFA

0x00000012 : 0xE7

Pic7



9. (5 points) Consider the following MIPS assembly code shown in the **figure below** (where R0, R1, and R2 are 32-bit general-purpose registers, and **R0** has initialized with a value of **0x00000002**). (a). (2 points) What is the value (hexadecimal) of the register **R0** after the execution of code. (b). (3 points) Show the **data memory addresses** (hexadecimal) of the instructions (ii) and (iv).

(a). (2 points) **R0 = 0x0000000C**

(b). (ii) (1.5 points) **0x00000006**

(iii) (1.5 points) **0x0000000F**

Pic8

- i) **ADDI R0, R0, 0x0001**
- ii) **LW R1, 0x0003 (R0)**
- iii) **SLL R0, R0, 0x0002**
- iv) **SW R2, 0x0003 (R0)**

10. (2 points) The source code of a high-level language compiler running on a MIPS system is shown below. Show its complete assembly code (where **g** and **h** are integer variables). *

Pic9

```
if (g != h)
    g = g + h;
else
    g = g - h;
```

```
LW R1, address g // address of g is not available
LW R2, address h // address of h is not available
BEQ R1, R2, SUBTRACT // BE means if R1 == R2 ,
ADD R1, R1, R2
J EXIT
SUBTRACT: SUB R1, R1, R2
EXIT:
```

11. (3 points) Based on the MIPS instruction sequence in the following figure, determine the values (in hexadecimal) of the registers **R1**, **R2**, and **R3** after the instruction execution (assume that the register **R0** is initialized with **0x00000002**). *

Pic10

```
1.  ADDI    R1, R0, 0x0005
2.  SLL     R1, R1, 1
3.  ADDI    R2, R0, 0x000A
4.  BEQ     R1, R2, NEXT
5.  SUBI    R1, R0, 8
6.  ADD     R3, R2, R0
7.  NEXT:  ADDI R3, R2, 0x0002
```

```
R1 = 0xFFFFFFFF
R2 = 0x0000000C
R3 = 0x0000000E
```

12. (1 point) Show an instruction that bypass the MEM stage of the MIPS system? *

ADD R2, R3, R4

13. (1 point) Consider an **S-stage** pipeline system executing a program with **K** instructions. Show that the pipeline speedup equals **K** when **S** goes to infinity. *

Speedup = $SK/(S+K-1)$, After dividing both numerator and denominator by S: $K/(1 + (K-1)/S)$, if $S \rightarrow \text{infinity}$, Speedup = K

14. (4 points) Find the answer of the question that is shown in the figure below: *

Avg.Inst.Exe.time(un-pipeline) = **9.7ns**, Avg.Inst.Exe.time(pipeline) = **2.4ns**.
Speedup = $9.7\text{ns}/2.4\text{ns} = 4.042$ times

Pic11

Consider an **unpipelined** processor. It has a **2 ns** clock cycle and uses **4 cycles for ALU operations, 5 for branches, and 6 for memory operations**. Assume that the relative frequencies of these operations are **50%, 15%, and 35%**, respectively. **Pipelining** the processor adds **0.4 ns** of clock overhead (ignore other pipeline latencies and hazard delays). Find out how much speedup the instruction execution rate will gain from the pipeline.

16. (2 points) Find the answer of the question that is shown in the Figure below: *

Speedup = no.of stages/ (ideal CPI + pipeline stall) = $5/(1+0.78) = 5/1.78 = 2.1$ times

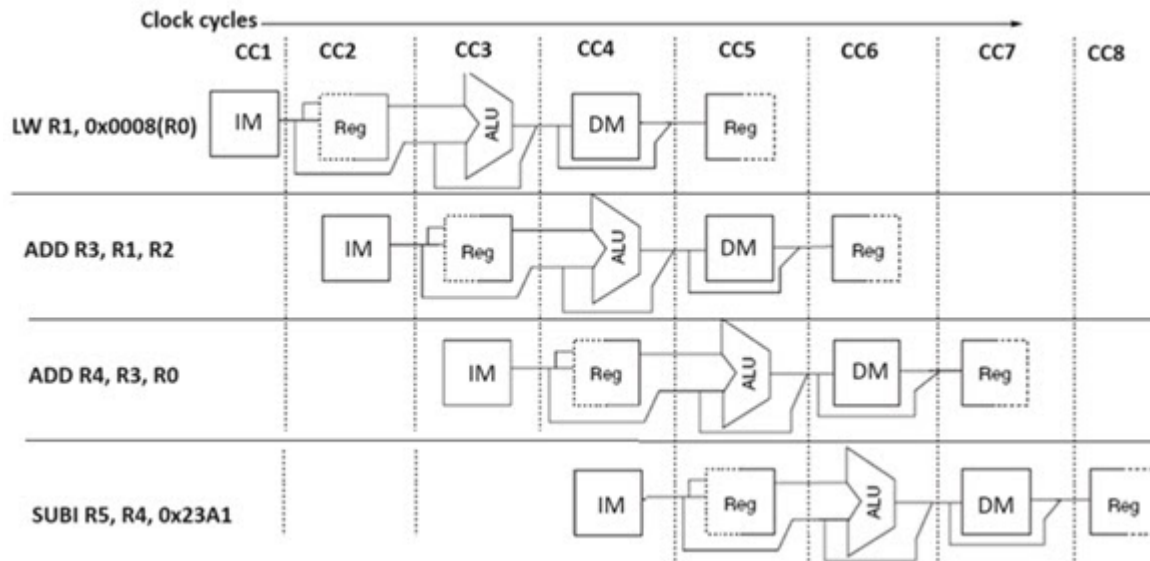
Pic12

Assume that the MIPS system with a clock speed **0.4 ns** has imposed a stall of **0.78** clock cycles per instruction for a particular user process. What is the **speedup** of the system?

17. (2 points) Consider the **pipeline data path** of the MIPS system shown in the **Figure** below. Is there any structural hazard in the system? If yes, describe the possible ways (if any) to eliminate it. *

There is no structural hazard in the system (different memories are in IF and MEM stages!)

Pic13



18. (6 points) Answer the following based on the MIPS instruction sequence shown in the figure below: (a). (2 points) Give the **line numbers of the instructions** and the **data-hazard-causing registers** (if any). (b). (4 points) If any **forwarding** or **stall** requires, then give the corresponding instruction's **stage name** and **clock cycle** that are supported for forwarding or stall. *

	1	2	3	4	5	6	7	8	9	10	11	12
1. SUBI R2, R1, 0x98AF	IF	ID	EX	MEM	WB							
2. AND R4, R5, R2		IF	ID	EX	MEM	WB						
3. ADD R3, R4, R7			IF	ID	EX	MEM	WB					
4. LW R8, 0x3420(R3)				IF	ID	EX	MEM	WB				
5. OR R9, R8, R6					IF	ID	STALL	EX	MEM	WB		
6. SW R9, 0x4690(R8)						IF	STALL	ID	EX	MEM	WB	
7. SUBI R6, R9, R4							STALL	IF	ID	EX	MEM	WB

Pic14

1. SUBI R2, R1, 0x98AF
2. AND R4, R5, R2
3. ADD R3, R4, R7
4. LW R8, 0x3420(R3)
5. OR R9, R8, R6
6. SW R9, 0x4690(R8)
7. SUBI R6, R9, R4

18. (3 points) Based on the given MIPS's **instruction memory (IM)**, answer the following: (a). (1 point) What is the **Program Counter (PC)** value if the **IF stage** selects the instruction **0x5B4C6732**? (b). (1 point) Show the value of the **PC** when the previous instruction was fetched. (c). (1 point) Suppose the value of the **PC** is **0x0000000D**. Show the fetched instruction. *

(a) . (1 point) 0x0000000C (b). (1 point) 0x00000008 (c). (1 point) 0x4C6732AF

Pic15

Memory Address	Memory Word
0x00000008	0x23
0x00000009	0x5A
0x0000000A	0x1E
0x0000000B	0xA0
0x0000000C	0x5B
0x0000000D	0x4C
0x0000000E	0x67
0x0000000F	0x32
0x00000010	0xAF

19. (2 points) Consider the IM memory in the above figure. If the current value of PC is 0x0000000A, then what will be the next value of PC? *

0x0000000E

20. (8 points) Consider the following **high-level code running on a 32-bit MIPS system**, shown below. **Show its MIPS assembly code translation, including complete memory operations with the data memory (DM) for the variable allocation.** Each variable's integer value is **32 bits**. The **32-bit memory address** of the first byte of the variable, **a**, is **0xA500C004**. The variables **b** and **c** use the consecutive byte addresses for their values in **DM**. Use the 32-bit registers **R1**, **R2**, and **R3** for **a**, **b**, and **c** in memory and arithmetic operations, and register **R0** as the base register, initialized to **0xA5000002**. *

MIPS instructions for the high-level code:

1. **LW R1, 0xC002(R0)** ; **R1 = 0x0000001E (30)**
2. **ADDI R2, R2, 0x000F** ; **R2 = 0x0000000F (15)**
3. **SUB R2, R2, R1** ; **R2 = 0x000000F1 (-15)**
4. **SW R2, 0xC006(R0)**
5. **ADDI R3, R2, 0x000A** ; **R3 = 0xFFFFFFFFB (-5)**
6. **SW R3, 0xC00A(R0)**

The DM is utilized for variable storage:

Pic16

```
int a = 30;
int b = 15 - a;
int c = b + 10;
```

32-bit Address	Data Byte	
0xA500C004	0x00	int a (32-bit) = 0x0000001E (30)
0xA500C005	0x00	
0xA500C006	0x00	
0xA500C007	0x1E	
0xA500C008	0xFF	int b (32-bit) = 0xFFFFFFFF1 (-15)
0xA500C009	0xFF	
0xA500C00A	0xFF	
0xA500C00B	0xF1	
0xA500C00C	0xFF	int c (32-bit) = 0xFFFFFFFFB (-5)
0xA500C00D	0xFF	
0xA500C00E	0xFF	
0xA500C00F	0xFB	

This content is neither created nor endorsed by Google.

Google Forms

